Answer key Tutors only

The numbers the web pages show. "Say" = the one point to make at the debrief. Tutor notes on page 3.

A SUV from the DICOM header

Real data: PETCT_7faf36a152, the hospital's own reconstruction

1 · THE IMAGE

1 **Right lower lobe, posterior, a few cm above the liver dome.**

2 **Bigger. With 0-5, everything above SUV 5 is full black, so the warm edge joins the tumour.**
Say: The window changes what you see, not the data. Remember this when you draw in Block D.

3 **SUVmax 12.49 (the hottest voxel). SUVmean depends on the circle: about 8.6 for a 2 cm circle, 6.1 for 3 cm, 4.4 for 4 cm.**
Say: SUVmax is one voxel; SUVmean is only as good as the circle someone drew.

4 **Heart muscle, kidneys and urine (renal pelvis, bladder), some gut. Normal FDG uptake and excretion, not cancer. (The brain is not shown: the data provider blanked the head.)**

2 · THE DICOM HEADER

1 **Weight 70 kg · dose 323 MBq · injection 07:44:00 · half-life 6586.2 s = 109.8 min · Series Time 08:39:01.**

2 **Series Time. Decay Correction = START: the scanner has already corrected every slice back to the series start. Using the slice's Acquisition Time (08:45:35, 6.6 min later) corrects twice: SUVmax 4 % too high.**

3 **Uptake `55.0 min` · fraction left `0.707` · dose at scan start `228 MBq` · SUVmax `40 713 ÷ (228 × 10⁶ ÷ 70 000) = 12.49`**

4 **BMI 24.8 · LBM `43.8 kg` · SULmax `12.49 × 43.8 ÷ 70 = 7.8`**

3 · WHAT IF?

Mistake	SUVmax	Change	Why
Weight in pounds (154)	27.47	+120 %	SUV is proportional to the weight.
Clock 30 min slow (07:14)	15.09	+21 %	85 min of decay assumed, so too little dose left in the calculation.
10 % left in the syringe	13.88	+11 %	The screen (12.49) is too low: she got less dose.

4 **No. The calculation removes the decay exactly, but not the biology: FDG keeps building up in most tumours, so a later scan reads higher. EANM: 60 min (55-75); a repeat scan within 10 min of the first uptake time.**

5 **Weight, dose (and what was left in the syringe), injection and scan times (clocks in sync), uptake time. Also the same scanner and reconstruction (Block C). Blood glucose is not in the DICOM header: ask for it.**
Say: Even with everything matched, SUVmax repeats only to about 12 %. QIBA: a real change from +39 % or −28 %.

B Attenuation correction

Simulated scan · 3 iterations, 5 mm filter

1 · WITH AND WITHOUT AC

1 **Too bright: skin and the body surface, the lungs. Too dark: the centre of the body (mediastinum, spine, middle of the liver).**

2 **About 3.9 with AC, 4.3 without (tumour max ÷ liver mean, 3 cm circle). The tumour stands out a little more without AC, because the deep liver is too dark. Pairs' circles give slightly different numbers.**
Say: Looking better is not measuring right. Without AC the contrast is a distortion.

3–4 **Both photons must leave the body. Lines through the middle cross about 30 cm of tissue: only about 1 pair in 18 gets out. Lines through the skin or air-filled lung lose little. No SUV without AC: the values have no unit.**

2 · BREATHING

CT used for AC	Tumour SUVmax	Change
2 cm higher (breathed out)	8.58	−16 %
Matched	10.19	–
1 cm lower	10.19	0 %
2 cm lower	9.42	−8 %
4 cm lower (deep breath in)	6.65	−35 %

1 **Goes down, to 8.58 (−16 %).** The moved CT puts lung where the tumour is, so lines through it are corrected too little.
2 **A curved band at the liver dome, too hot or too cold; the tumour too low.**
3 **4 cm lower (deep breath in): −35 %.**

4 **On PET on CT: the liver dome and the diaphragm do not line up, a curved cold band under the lung, the PET lesion sitting off the CT mass. Look there every time.**

5 **Shallow breathing for the CT, not a deep breath in · an averaged CT · gated PET with a 4D CT.**
Say: The CT is in the PET. A breathing mismatch moves the SUV with nothing changed in the patient (Erdi 2004: up to 24-30 %).

Patient: FDG-PET-CT-Lesions (autoPET), PETCT_7faf36a152 (Gatidis et al., Sci Data 2022;9:601; TCIA, doi:10.7937/gkr0-xv29; CC BY 4.0). SUV: Boellaard et al., EJNMMI 2015;42:328-54. QIBA FDG-PET/CT profile v1.14 (2023). LBM: Janmahasatian et al., Clin Pharmacokinet 2005;44:1051-65.

C Reconstruction and lesion size

Simulated · OSEM, 16 subsets

1–2 **SUVmax up, liver mean the same, noise up.**

No filter	1 iteration	3 iterations	12 iterations
Tumour SUVmax	11.34	12.43	13.05
Liver mean	2.61	2.65	2.64
Liver noise (SD, % of mean)	15 %	23 %	44 %

3 **3 iterations: no filter 12.43 · 3 mm 12.01 · 5 mm 10.19 · 7 mm 9.10. Noise 23 → 21 → 13 → 11 %. The filter smooths: less noise, but a lower SUVmax.**

4 **None of them. 9.1 to 13.1 here, 12.49 in the hospital (sharper reconstruction: resolution modelling, time of flight, 2 mm pixels).**

Say: There is no one right SUVmax. Compare SUVs only between scans reconstructed the same way.

5 **Same scanner type and reconstruction? Both EARL-accredited, with the EARL reconstruction? Same uptake time and protocol?**

1 **Small ones lower.**

2 **3 it + 5 mm: 10 mm 36 %, 37 mm 103 %.**

3 **From about 17-22 mm (89-96 %).** Below 2-3 × the blur (6.5 mm) they read low.

4 **Not only one: 5 mm filter with 3 to 12 iterations, and 2 iterations with 3 mm.** No filter, or 7 mm: fails.

5 **About 8** (3 ÷ 0.36), if it is a sphere of even uptake. It could be anywhere above 3.

Sphere	Recovery	EARL range
10 mm	36 %	34-57 %
13 mm	60 %	59-85 %
17 mm	89 %	73-101 %
22 mm	96 %	83-109 %
28 mm	106 %	91-113 %
37 mm	103 %	95-116 %

Say: Small lesions read low. A tumour that shrinks reads a lower SUVmax even if each cell takes up as much FDG.

D Contouring on PET

3 iterations, 5 mm filter · expert outline 25.4 mL

1.1–3 **Pairs' own numbers.** A wide window (0-5) gives a bigger outline, a narrow one (0-20) a smaller one.

2.1 **Biggest: SUV 2.5 · smallest: gradient.**

2.3 **SUV 2.5 overlaps the expert best (0.75)**, only because her outline is big and her uptake uneven (a hot core off-centre). It is not a sign that SUV 2.5 is good.

Method	Threshold	mL	Overlap
41 % of SUVmax	4.18	7.8	0.47
50 % of SUVmax	5.09	5.1	0.33
SUV 2.5	2.50	15.1	0.75
Adaptive	3.66	9.2	0.53
Gradient	–	3.8	0.26

2.4 **It grows into anything warm next to the tumour. In the liver (background about 2.5) or the brain it fails completely.**

2.5 **No single method. In lung the GTV is contoured on CT, with PET as a guide; an automatic outline is a start that a physician edits.**

3 · ANOTHER RECONSTRUCTION

Reconstruction	SUVmax	41 % mL	SUV 2.5 mL
1 iteration, 7 mm	8.48	8.6	14.4
3 iterations, 5 mm	10.19	7.8	15.1
12 iterations, none	13.05	5.5	15.4

1 **Smaller (7.8 → 5.5 mL).**

3 **The threshold follows the maximum:** a sharper, noisier image → higher SUVmax → 41 % of it is higher → the outline shrinks.

4 **17-37 mm within 10 %.** 13 mm +90 %, 10 mm +394 %: small spheres need higher thresholds.

Say: Change the reconstruction, change the GTV. A PET contour comes with its scanner, its reconstruction and its method.

Methods and SUV 2.5: Hatt et al., AAPM TG211, Med Phys 2017;44:e1-e42. EARL1 ranges: Kaalep et al., Eur J Nucl Med Mol Imaging 2018;45:1344-61. Partial volume: Soret et al., J Nucl Med 2007;48:932-45. Breathing: Erdi et al., J Nucl Med 2004;45:1287-92; Goerres et al., Radiology 2003;226:906-10; Liu et al., Phys Med Biol 2009;54:7345-62.

Checklist · suggested answers

1 · SUV calculated right?	Weight, dose (minus what was left), injection and scan times with synced clocks, uptake time 55–75 min and matched on repeat scans.
2 · Does the CT match the PET?	Look at PET on CT at the liver dome and around the tumour. Shallow breathing for the CT; no SUV without AC.
3 · Reconstruction and lesion size?	Same scanner and reconstruction (EARL) on both scans. Lesions under about 2 cm read low.
4 · Which contouring method?	Which method, threshold and window, on which reconstruction? Check and edit; in lung, contour the GTV on CT.

Tutor notes For René

Everything runs in the browser. Nothing to install, no login.

Before the session

- Open **patrikbrynolfsson.github.io/estro2026-dwi-handson/pet.html** on the room's PC and on your laptop. The QR code on the worksheet goes there.
- The first load takes a few seconds (about 9 MB of images). On the room's Wi-Fi, ask pairs to open it at the start.
- A page that looks broken: Ctrl+F5 once (old copy in the browser).
- Print one worksheet per pair, double-sided. Pairs can also type their answers on the pages.

How the pages work

- One task at a time: Previous / Next under the task title. Predict boxes lock once filled in; then pairs try it.
- Answers stay in that browser only; nothing is sent anywhere. "My answers" (top of each page) lists them, with links back to each task.
- Each block ends with a message box: that is the take-home point.
- Laptops are best; phones work but are cramped.

Time	Block	Min	Take-home point
15:00	Intro: "When you're shown an SUV or a PET contour, what do you need to know before you trust it?"	10	The question
15:10	A · SUV from the DICOM header	20	SUV is a calculation
15:30	B · Attenuation correction	20	The CT is in the PET
15:50	C · Reconstruction and lesion size	25	Same patient, different SUVmax
16:15	Break	10	
16:25	D · Contouring on PET	25	The method determines the GTV
16:50	Debrief: go through the worksheet; fill in the checklist together	10	The checklist

If time slips

- Shorten B (skip B 1.3–1.4 and B 2.4–2.5) and the phantom part of C (C 2.4–2.5).
- **Protect D, and its step 3** (another reconstruction): it ties C to the GTV.
- Fast pairs: the bonus tasks (marked on the pages) and the "What if?" step in A.

Questions you may get

- **Real or simulated?** Block A is her real scan and DICOM header. B–D are a simulated scan built from her real PET and CT, so we can change one thing at a time.
- **Why 10.19 in B-D but 12.49 in A?** A is the hospital's sharper reconstruction; B–D use a standard one. That is Block C's point.
- **Is the expert outline a GTV?** No: the lesion outline from the autoPET dataset, drawn for research, not for radiotherapy.
- **RayStation?** The programme mentions it; these pages replace it, with the same patient case.

Built by Patrik Brynolfsson for ESTRO Advanced Imaging 2026. Data: FDG-PET-CT-Lesions (autoPET), TCIA, doi:10.7937/gkr0-xv29 (CC BY 4.0). Simulation: the patient's CT → attenuation at 511 keV (Carney et al., Med Phys 2006;33:976–83), blur 6.5 mm, Poisson noise, OSEM with 16 subsets. Numbers read from the web pages on 2 November 2026.